Advancing Segment Anything Model for Efficient Salient Object Detection in Remote Sensing Images

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***Abstract*—Salient object detection in optical remote sensing images (ORSIs-SOD) often relies on leveraging pretrained knowledge from natural images to achieve high accuracy with limited training data. Traditional methods typically employ vision backbones [e.g., convolutional neural networks (CNNs) or vision transformers (ViTs)] pretrained on ImageNet to extract features from ORSI scenes. However, these backbones exhibit limited generalization across diverse scenarios compared to recent vision foundation models. To this end, we propose ORSI-segment anything model (SAM), a novel ORSI-SOD framework based on the SAM, leveraging its superior generalization capabilities to achieve an exceptional efficiency–accuracy tradeoff. Specifically, ORSI-SAM adopts lightweight SAM as the backbone, effectively reducing parameter size and computational overhead to enable efficient deployment on satellite devices while retaining the rich knowledge learned from large-scale natural image datasets. To mitigate the impact of unavailable prompts in ORSI-SOD on the prediction capability of the SAM decoder, we introduce a hierarchical interaction prompt generator (HIPG), which aggregates hierarchical features and generates mask prompts tailored for salient objects to guide the decoder in producing high-quality saliency maps. Furthermore, to address the recognition challenges caused by the inherent characteristics of ORSIs, we propose a semantic-aware refinement decoder (SARD). SARD integrates structural details from low-level features to enrich fine-grained object information while leveraging high-level features to suppress redundant interference in shallow layers, thereby improving the detailed information in the predicted saliency map. ORSI-SAM is the first work to explore the accuracy–efficiency tradeoffs for ORSI-SOD based on SAM architecture. Extensive experiments on benchmark datasets show that ORSI-SAM achieves superior performance compared to recent state-of-the-art methods with 12.2 M parameters and 8.9 G FLOPs.**

***Index Terms*—Lightweight salient object detection (SOD), optical remote sensing image, segment anything model (SAM).**

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I. Introduction

SALIENT object detection for optical remote sensing images (ORSIs-SOD) aims to precisely localize perceptually salient regions (e.g., artificial structures, road networks, and river systems) from large-scale ORSIs. By efficiently extracting salient regions, ORSI-SOD significantly reduces redundant computation and enhances the interpretation efficiency of large-scale remote sensing imagery, demonstrating critical value for various remote sensing applications, such as land use [1], urban development [2], and disaster monitoring [3]. Although recent years have witnessed remarkable progress, current ORSI-SOD methods still face fundamental challenges that limit their deployment on resource-constrained satellite platforms.

Complex Scene Understanding. Modern remote sensing imaging systems capture wide-area and high-angle scenes with extreme complexity [4]. Redundant features of backgrounds (vegetation, cloud cover, and shadows) suppress object saliency, degrading model discriminability. In addition, objects in ORSIs exhibit significant category diversity, extreme scale variations, and sparse distributions, especially numerous small targets that pose detection challenges. These characteristics pose fundamental challenges for high-accuracy SOD.

Limited Annotated ORSI Training Data. The high acquisition and annotation costs of remote sensing images severely constrain the scale of training datasets, making it difficult to capture the diversity of real-world scenarios and objects, consequently degrading the performance of the model. Although current approaches commonly employ transfer of knowledge from natural image datasets (e.g., ImageNet-1K) to improve generalization, these methods still struggle to bridge the substantial domain gap between natural and remote sensing imagery.

Suboptimal Accuracy–Efficiency Tradeoff. Achieving an optimal accuracy–efficiency tradeoff remains a critical challenge for deploying ORSI-SOD models on resource-constrained platforms (e.g., satellites and edge devices). Although state-of-the-art methods deliver high detection accuracy, they typically rely on computationally intensive architectures, rendering them impractical for real-time processing.

Therefore, in complex ORSI scenes, how to effectively model salient features to distinguish foreground from cluttered backgrounds? how to enhance model generalization across diverse ORSI scenarios under limited training data constraints? and how to optimally balance detection accuracy with com-

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putational efficiency? Exploring these questions is crucial to achieving practical real-time ORSI-SOD systems.

Recent advancements in the ORSI-SOD task leverage convolutional neural networks (CNNs) and vision transformers (ViTs) pretrained on ImageNet to enhance their ability to capture semantic features of diverse objects in complex remote sensing scenarios, particularly under limited training data conditions. Specifically, CNN- and ViT-based ORSI-SOD approaches [5], [6], [7], [8], [9], [10], [11] commonly adopt pretrained backbones such as VGG [12], PVT [13], and Swin Transformer [14] as feature encoders. These approaches further incorporate specifically designed strategies tailored for ORSI-SOD to address the unique challenges posed by ORSIs, including edge guidance [5], feature enhancement [6], [10], and multiscale feature aggregation [7], [8], which effectively improve the model's capability to distinguish foreground from background pixels. The pretrained knowledge from ImageNet-1K, which comprises one million training samples, significantly enhances model representative capabilities, addressing the challenges associated with the limited availability of annotated ORSI data. However, traditional backbones derived from ImageNet-1K exhibit limited generalization across diverse and complex scenarios when compared to recent vision foundation models [15], [16], [17]. To further improve performance in the ORSI-SOD task, this study explores leveraging vision foundation models pretrained on large-scale natural image datasets, such as the SA-1B dataset, which contains over 11 million images, to better capture the diverse and intricate characteristics of ORSIs.

The segment anything model (SAM) [15] has gained prominence as a foundational vision model due to its remarkable generalization capabilities across a wide range of downstream segmentation tasks [18], [19], [20], [21]. SAM has demonstrated remarkable adaptability to pixel-level segmentation tasks in diverse scenarios with limited training data. However, the original SAM's heavyweight image encoder prohibits real-time execution on resource-constrained devices. Recent approaches [19], [22], [23] have developed lightweight variants of SAM by distillation. Among these methods, IRSAM [19] and SimIRSTD [23] illustrate that lightweight SAM architectures can maintain cross-domain generalization capabilities. To this end, we employ MobileSAM [22] as the backbone for benefiting from large-scale knowledge while maintaining high efficiency. In the latest work, MDSAM [18] introduced the first SAM-based architecture specifically designed for the SOD task in natural scene images (NSI-SOD), effectively demonstrating the potential of SAM to improve the perception of salient objects. The application of these designs to ORSI-SOD has two constraints listed as follows.

1) *Domain Gap:* As shown in Fig. 1, ORSIs typically contain numerous sparsely distributed small targets and complex background interference. These characteristics frequently lead to information loss of salient objects and identification challenges, significantly constraining the transfer efficacy of foundation model knowledge to ORSI-SOD tasks.

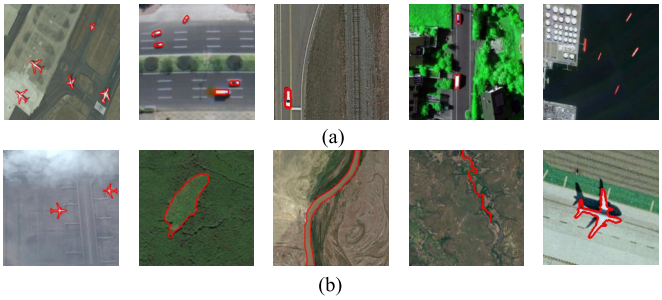


Fig. 1. Domain gap illustration in ORSI-SOD. (a) Small objects with a sparse distribution in ORSIs and (b) feature response of objects is suppressed by complex backgrounds in ORSIs. The red contours indicate the annotated salient objects.

2) *Task Gap:* As shown in Fig. 2, the original SAM relies on external prompts (points, boxes, or coarse masks) to generate class-agnostic predictions. However, such prompt information is typically unavailable during the inference stage. MDSAM adapts SAM through encoder–decoder fine-tuning, which critically limits the lightweight decoder's capacity to produce high-quality saliency masks.

To address these challenges, we introduce two key modules: the hierarchical interaction prompt generator (HIPG) and the semantic-aware refinement decoder (SARD). Specifically, HIPG consists of the lightweight multilevel feature fusion module (LMFFM) and the salient object prompt generator (SOPG) to efficiently aggregate hierarchical semantic features from different layers and generate a learnable mask prompt for the decoder. HIPG mitigates information degradation to improve the representation of the image embedding and the learnable prompt, facilitating the quality of the saliency map generated by the decoder. SARD promotes interaction between shallow and deep features by structure-aware spatial modulation (SSM), effectively filtering redundant background information from the shallow layers while emphasizing the structural details of salient objects. This approach significantly improves scene understanding and enriches the detailed representation of saliency maps. By combining these components, we propose the first lightweight SAM-based ORSI-SOD architecture, termed ORSI-SAM, to achieve an exceptional efficiency–accuracy tradeoff.

Our contributions can be summarized as follows.

1) We propose ORSI-SAM, the first ORSI-SOD framework based on lightweight SAM, achieving an exceptional efficiency–accuracy tradeoff. ORSI-SAM is a domain-specialized framework that strategically integrates remote sensing priors and SOD-specific designs to systematically mitigate both task and domain gaps for enhancing SAM's generalization capability for ORSI-SOD.
2) We propose the HIPG and SARD to mitigate the aforementioned challenges. HIPG dynamically generates learnable mask prompts of salient objects through multilevel feature aggregation to reduce information loss of diverse targets and improve decoder localization ability.

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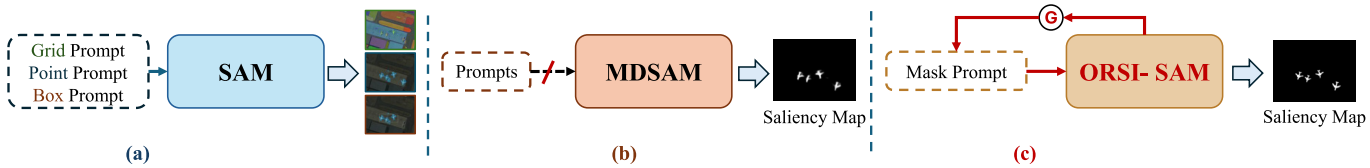


Fig. 2. Task gap between SAM and SOD task. (a) SAM [15] requires grid, point, and box prompts to generate a predicted mask. (b) MDSAM [18] removes prompts and only fine-tunes encoder–decoder. (c) Our ORSI-SAM incorporates with learnable mask prompt for generating a saliency mask. **G** denotes the prompt generator. ORSI-SAM effectively utilizes the prompt mechanism in SAM to improve the quality of decoder generated masks.

SARD refines the boundary cues of salient objects and mitigates interference of redundant background information in shallow layers, facilitating the generation of high-quality saliency maps. These two modules significantly boost ORSI-SOD performance with minimal computational overhead.

3) We evaluate the proposed ORSI-SAM on three publicly available datasets: ORSSD, EORSSD, and ORSI-4199. The extensive experiments demonstrate the effectiveness of our design. The proposed ORSI-SAM outperforms previous ORSI-SOD methods using only 12.2 M parameters and 8.9 G FLOPs. Compared to the latest SAM-based SOD model MDSAM, our method reduces 87.9% of the parameter count and 80.8% of the computational complexity and achieves superior performance.

The rest of this article is organized as follows. Section I-C introduces the related work on ORSI-SOD methods and SOD methods based on SAM. Then, our proposed ORSI-SAM is presented in Section III with detailed illustrations and discussions. In Section IV, we validate the effectiveness of our methods via comparative experiments and conduct ablation studies to further investigate our designs. Section V provides the limitations and future work of the proposed ORSI-SAM. Finally, this article is concluded in Section VI.

II. Related Work

A. ORSI SOD

SOD plays a vital role as an essential preprocessing step in various visual tasks and has been extensively investigated across diverse domains, including NSIs [5], [24], [25], [26], videos [27], [28], and RGB-D data [29], [30]. For ORSIs [31], SOD holds significant potential for interpreting critical scene information and facilitating detailed analyses in various remote sensing applications, such as disaster monitoring and urban development. Compared to NSIs, ORSIs cover expansive areas captured from high shooting angles and exhibit significant diversity in object types, colors, scales, and shapes. Complex background interference suppresses the intrinsic features of objects, posing the challenge for the model to efficiently distinguish foreground and background pixels in diverse scenes.

To address these challenges, numerous studies focused on advancing SOD methods specifically designed for ORSIs, referred to as ORSI-SOD [4], [5], [8], [9], [11], [32], [33], [34], [35], [36]. As a pioneering work, LVNet [9] proposed a two-stream pyramid module to perceive the diverse scales and local details of salient objects. PDFNet [4] explored multilevel feature interactions for ORSI-SOD. SARNet [37] roughly located salient objects in the semantically guided decoder and then refined the predictions by multilevel features. RRNet [8] designed a relational reasoning encoder to infer the semantic relationship in salient objects. ACCoNet [11] explored the coordination of adjacent features to capture contextual information. SFANet [35] utilized progressive semantic flow and uncertainty-aware refinement to generate an accurate prediction. HFCNet [36] designed a new hybrid heterogeneous encoder that combines CNN and ViT to model local and global information. Furthermore, to facilitate efficient deployment on resource-constrained satellites, CorrNet [6] lightened the VGG-16 for efficient feature extraction and adopted lightweight refinement modules to progressively generate saliency maps. SeaNet [10], which adopted MobileNet-V2 as its feature encoder, introduced a dynamic semantic matching module and an edge self-alignment module to enhance feature representation.

The superior performance of these ORSI-SOD methods is primarily attributed to leveraging CNN or ViT models pretrained on NSIs as encoders, which significantly enhances their capability to capture scene information under limited training data. However, the models traditionally pretrained on ImageNet exhibit weaker generalization in diverse scenarios compared to recent vision foundation models, such as CLIP [16], DINO [17], and SAM [15]. In this study, we investigate the adaptability of SAM, a foundational model pretrained on the largest segmentation dataset, to ORSIs, aiming to enhance the performance of ORSI-SOD tasks.

B. SAM for SOD

Recently, SAM has been recognized as a vision foundation model due to its exceptional generalization capability across various downstream segmentation tasks [18], [19], [20], [21], [38], [39], [40]. In SOD tasks, the precise prompts required by SAM, such as points, bounding boxes, or masks, are often challenging to acquire. Some studies have fully fine-tuned SAM to adapt it for SOD tasks. However, this approach can result in an excessive number of trainable parameters and may even lead to performance degradation. To address these challenges, alternative approaches have been proposed to adapt SAM with a minimal number of trainable parameters. For instance, SSOM [41] employs low-rank adaptation (LoRA) to fine-tune SAM for SOD tasks. Similarly, HQ-SAM [42] introduces a learnable high-quality output token injected into SAM's mask decoder to enhance the prediction of high-quality masks. MDSAM [18] further innovates with a lightweight multiscale adapter (LMSA) and detail-enhancing modules

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to enable SAM to learn multiscale and multilevel features, significantly improving the generalization ability of SAM on SOD tasks.

However, the excessive number of parameters and high computational complexity of these SAM-based methods present significant challenges for deploying them on resource-constrained satellites to perform ORSI-SOD tasks effectively. Besides, the original design of SAM relies on external prompts such as points, boxes, or rough masks to generate category-independent segmentation predictions. However, in the SOD task, such prompts are not obtainable in the inference. The significant difference in task settings significantly limits the ability of lightweight decoders to generate high-quality saliency masks. MDSAM [18] removes the prompt encoder and only fine-tunes the encoder and decoder, which limits the decoder's predictive ability. In this work, this study explores the application of lightweight SAM in the ORSI-SOD task, aiming to achieve hardware-friendly deployment while preserving the pretrained knowledge derived from large-scale natural image datasets. Furthermore, SAM-based architectures tailored for the ORSI-SOD task require further exploration.

C. SAM in Remote Sensing Images

Building upon the SAM's demonstrated capabilities in generalization and zero-shot learning, recent studies have explored the integration of SAM into remote sensing scenarios, allowing efficient training with limited annotations while maintaining strong adaptability across diverse geographical scenarios and multimodal data.

RingMo-SAM [20] builds a large-scale dataset with millions of segmentation instances collected from multiple open-source remote sensing images for multimodal image segmentation, which can not only segment anything in ORSIs and synthetic aperture radar (SAR) images [43], [44] but also identify object categories. RSPrompter [45] introduces novel prompt learning to facilitate the ability of SAM for remote sensing image instance segmentation. PointSAM [46] leverages point annotations to fine-tune SAM for remote sensing images and achieves comparable performance with fully supervised methods. For infrared images, previous works [47], [48], [49], [50] adopt the CNN or transformer backbone to achieve high performance on infrared small target detection (IRSTD). IRSAM [19] narrows the notable domain gap between natural and infrared images and achieves a superior performance of IRSTD based on SAM. MAF-SAM [51] effectively utilizes SAM to extract information from multispectral remote sensing images for crop monitoring.

Extensive advancements validate SAM's remarkable cross-modal generalization capability, demonstrating consistent performance across heterogeneous remote sensing data modalities, including infrared and SAR images. Motivated by these findings and the success of SAM in the generic SOD task, we systematically investigate optimization strategies to enhance its performance for the ORSI-SOD task. Our work reveals the potential of SAM for advancing the interpretation of complex remote sensing scenes.

III. Methods

In this section, we introduce the proposed ORSI-SAM for the ORSI-SOD task. The overall architecture can be observed in Fig. 3. The ORSI-SAM consists of three key components: backbone based on MobileSAM, HIPG, and SARD. We will first describe the overall architecture of our framework in Section III-A and then detail each component in Sections III-B and III-C.

A. Overall Architecture

In ORSI-SAM, we leverage a lightweight backbone from MobileSAM [22] to extract multilevel semantic features. This design effectively reduces the excessive parameter size and computational overhead of the original SAM while preserving the pretrained knowledge acquired from large-scale datasets. We denote the input image as $I \in \mathbb{R}^{3\times H\times W}$, where H and W denote the height and width of the input, respectively. At the beginning of the encoder, the patch embedding layer utilizes two convolution blocks with a stride of 2 to reduce the resolution. Then, the four stages of the encoder gradually process the embedding feature F_e. The first stage is constructed by convolution blocks with inverted residuals, while the remaining three stages are transformer blocks. The downsampling operation between different stages is processed by convolution blocks with a stride of 2. The output features of each stage are denoted as $F_i, i = \{1, 2, 3, 4\}$. To improve the adaptability of SAM and mitigate information degradation during the downsample operation, we utilize HIPG to aggregate low- and high-level information in F_e and F_{1-4} to generate multilevel representative image embedding feature F_f, deep feature F_d, and mask prompt S_g for the SAM mask decoder and SARD to generate high-quality saliency map. In particular, the upsampled S_g is then used to generate S_{mask} by the upsampling operation to calculate the loss with ground truth. Furthermore, to address weak object feature responses caused by background suppression, we propose SARD in conjunction with the original SAM mask decoder to model the local boundary cues for generating a saliency map with fine-grained details. The SARD utilizes high-level semantic information from F_d and the mask decoder feature F_m to eliminate redundant details and activate the representation of the structure of salient objects in the shallow feature F_{local}, effectively refining the detailed information of the generated saliency map S_f. The final saliency prediction S_{out} is obtained through the weighted fusion of S_f and the mask decoder output S_m. The training loss of ground truth guidance consists of two parts: the prediction S_{out} and the upsampled mask prompt S_{mask}. During the inference stage, S_{out} is utilized to obtain the final prediction.

B. Hierarchical Interaction Prompt Generator

To bridge the task gap between the original SAM and the ORSI-SOD task, we propose the HIPG, which enhances the adaptability of the SAM architecture to ORSI-SOD through hierarchical aggregation of features and the generation of learnable prompts. HIPG comprises two key components: 1) an LMFFM that progressively integrates multistage features

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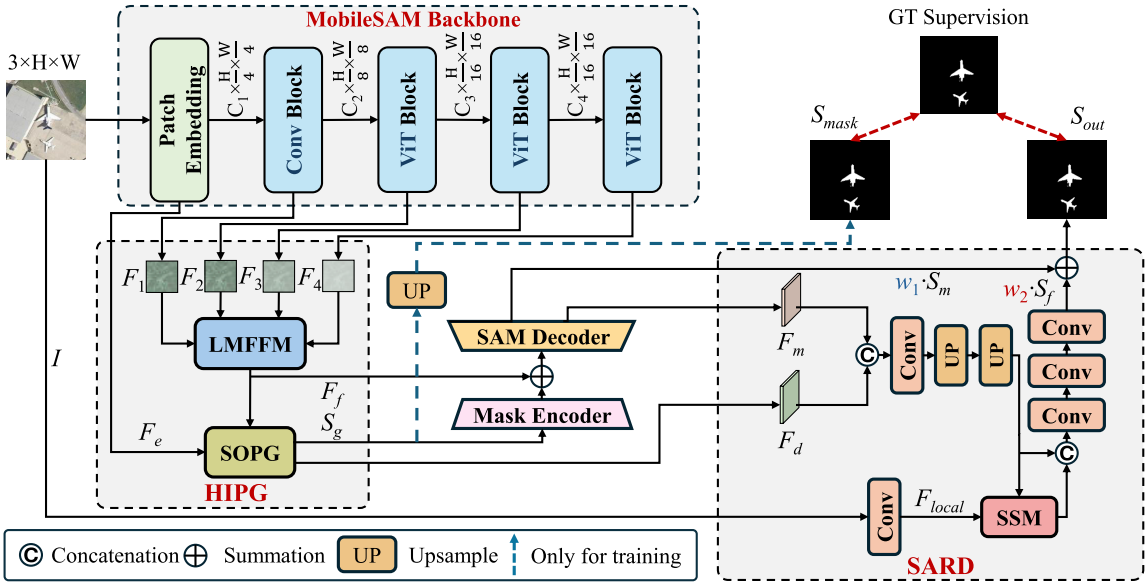


Fig. 3. Overall architecture of the proposed ORSI-SAM. ORSI-SAM consists of three key designs: efficient MobileSAM backbone, HIPG, and SARD. HIPG consists of two components: LMFFM and SOPG. SARD utilizes SSM to refine object response and suppress the interference of backgrounds.

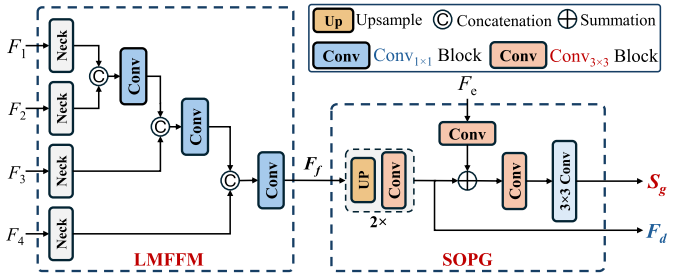


Fig. 4. Detailed information of the proposed HIPG. HIPG consists of LMFFM and SOPG to generate hierarchical image embedding F_f and learnable mask prompt S_g. F_d is obtained by performing 4× upsampling on F_f.

to refine multiscale representations for accurate mask prompt generation and decoder optimization and 2) an SOPG that leverages cross-level feature interactions to generate informative prompts, effectively preserving shallow-layer scene information and mitigating information loss from downsampling in MobileSAM, particularly crucial for small targets.

1) Lightweight Multilevel Feature Fusion Module: In the original MobileSAM, the mask decoder receives high-level image embedding features from the encoder to generate predictions. However, the loss of low-level detail information in the shallow layers makes it difficult for the model to accurately locate objects, especially small objects. To address this issue, we propose the LMFFM to efficiently integrate information across different layers. As illustrated in Fig. 4, LMFFM progressively fuses multilevel features F_{1-4} extracted from the MobileSAM encoder.

In LMFFM, we first apply a neck module to project the output feature F_i of each stage to $\hat{F}_i \in \mathbb{R}^{256\times(H/16)\times(W/16)}$ for dimension alignment. The neck module processes input F_i as Conv1 × 1 → LN → Conv3 × 3 → LN. Notably, in the neck module for F_1, the stride of Conv3 × 3 is set to 2 to reduce the spatial size to $(H/16)\times(W/16)$. Then, the fusion block $\Phi_{1\times1}(\cdot)$ is utilized to progressively aggregate the neighbor features by 1 × 1 convolution from shallow to deep layer, which can be formulated as follows:

$$\Phi_{1\times1}(x_1, x_2) = \mathrm{ReLU}(\mathrm{BN}(\mathrm{Conv1}\times1(\mathrm{Cat}[x_1, x_2]))) \quad (1)$$

$$F_f = \Phi_{1\times1}\left(\hat{F}_4, \Phi_{1\times1}\left(\hat{F}_3, \Phi_{1\times1}\left(\hat{F}_2, \hat{F}_1\right)\right)\right) \quad (2)$$

where Cat[·] denotes the catenation operation and F_f is the multilevel image embedding for the mask decoder. The LMFFM generates rich object information for the decoder, providing effective guidance for the SOPG to generate more accurate prompts.

2) Salient Object Prompt Generator: In both original and lightweight SAM architectures, accurate prompts (e.g., points, boxes, and masks) are essential for the decoder to produce high-quality segmentation maps. However, such prompts are typically unavailable in SOD tasks. A prior approach [18] bypasses the prompt encoder and directly inputs embedding features into the decoder, which constrains the decoder's capability to leverage pretrained knowledge from large datasets. To overcome this limitation, we propose the SOPG to generate learnable mask prompts tailored for the decoder. As illustrated in Fig. 4, SOPG integrates the patch embedding output F_e and the LMFFM output F_f to produce the mask prompt $S_g \in \mathbb{R}^{1\times(H/4)\times(W/4)}$ and deep features F_d. F_d is utilized to enhance the feature semantic representation in the decoder stage. The inclusion of F_e is particularly important as it retains more low-level information about salient objects.

In SOPG, the LMFFM output $F_f \in \mathbb{R}^{256\times(H/16)\times(W/16)}$ is passed through two cascaded upsampling and $\phi_{3\times3}$ blocks to obtain deep feature $F_d \in \mathbb{R}^{32\times(H/4)\times(W/4)}$, which can be

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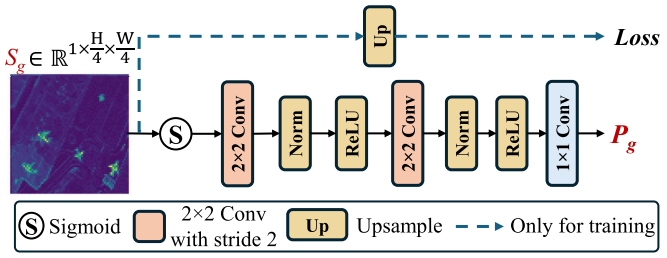


Fig. 5. Detailed information of MaskEncoder. The MaskEncoder utilized two 2×2 convoultions with stride of 2 to extract prompt feature P_g for image embedding.

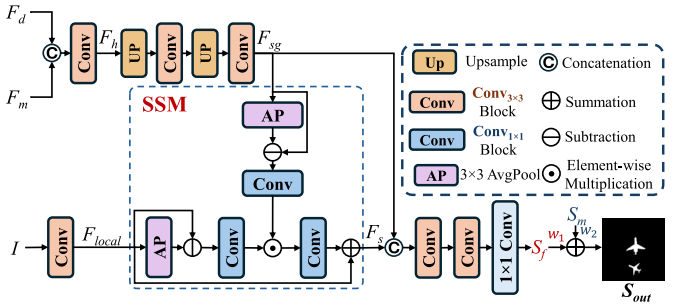


Fig. 6. Detailed information of the proposed SARD. SARD tasks image I, mask decoder feature F_m, and deep feature F_d as input and generates the final prediction S_{out}.

formulated as follows:

$$F_d = \phi_{3\times3}\left(\text{Up}_{\times2}\left(\phi_{3\times3}\left(\text{Up}_{\times2}\left(F_f\right)\right)\right)\right) \tag{3}$$

where $\text{Up}_{\times2}(\cdot)$ denotes the $2\times$ upsample operation by bilinear interpolation and $\phi_{3\times3}$ processes the input as Conv3 $\times$ 3 $\rightarrow$ BN $\rightarrow$ ReLU. Then, F_d and embedding feature F_e are aggregated for obtaining mask prompt S_g by two projection blocks ϕ_{p} and a 3×3 convolution, which can be formulated as follows:

$$S_g = \text{Conv3} \times 3\left(\phi_{\text{p}}\left(\phi_{\text{p}}\left(F_e\right) + F_d\right)\right) \tag{4}$$

where $\phi_{\text{p}}(\cdot)$ denotes the block to process the input as Conv3 $\times$ 3 $\rightarrow$ LN $\rightarrow$ ReLU and the final Conv3 $\times$ 3 reduces the feature channel to 1 for generating mask. S_g is regarded as a mask prompt for MaskEncoder and is also utilized to calculate the loss with ground truth.

Based on F_f and S_g obtained from HIPG, the image embedding for the mask decoder can be formulated as follows:

$$F_{\text{embed}} = \text{MaskEncoder}\left(S_g\right) + F_f \tag{5}$$

where MaskEncoder$(\cdot)$ is the mask prompt encoder. The architecture of MaskEncoder can be found in Fig. 5. S_g is first passed through a sigmoid function, and two 2×2 convolution blocks are utilized to align the spatial and channel dimensions between S_g and F_f. F_{embed} will be used as the image embedding for the mask decoder of MobileSAM to generate mask decoder feature F_m and mask decoder prediction S_m. Different from the SAM-based methods [22], [42], the output features of the proposed HIPG fully leverage low- and high-level information from the MobileSAM's encoder to mitigate information loss.

C. SARD

With HIPG, ORSI-SAM effectively mitigates performance degradation caused by the loss of object information. However, due to the imaging mechanism, object feature representation in ORSIs is often suppressed by background interference, leading to saliency maps that lack detailed object information. To address this limitation, we propose the SARD to refine object structure representation and alleviate the impact of background interference. The key ideas of SARD are: 1) enhancing structural responses of salient objects in shallow features to enrich fine-grained detail information and 2) leveraging deep semantic features as guidance signals to filter out redundant information in shallow layers. The detailed information of SARD can be observed in Fig. 6.

The input of SARD consists of three parts: the input image I, mask decoder feature F_m, and deep features F_d from HIPG. The deep features lack detail information on objects. Therefore, we first apply a 3×3 convolution block to extract the more local features, which can be formulated as follows:

$$F_{\text{local}} = \phi_{3\times3}\left(I\right). \tag{6}$$

Then, we concatenate the mask decoder feature F_m and deep features F_d as high-level semantic guidance. The process can be formulated as follows:

$$F_h = \phi_{3\times3}\left(\text{Cat}\left[F_d, F_m\right]\right) \tag{7}$$

$$F_{\text{sg}} = \phi_{3\times3}\left(\text{Up}_{\times2}\left(\phi_{3\times3}\left(\text{Up}_{\times2}\left(F_h\right)\right)\right)\right). \tag{8}$$

Since the shallow information F_{local} contains significantly redundant background features that can degrade prediction performance, further careful design of decoder is necessary. We regard $F_{\text{sg}} \in \mathbb{R}^{32\times H\times W}$ as guidance to highlight the foreground object region in the shallow feature $F_{\text{local}} \in \mathbb{R}^{32\times H\times W}$. To achieve this, we design an SSM to leverage boundary information as a signal to refine the structure response of salient objects. This module not only effectively filters out redundant background information from low-level features but also explicitly models local boundary structures, enhancing the decoder's sensitivity to target boundary characteristics. In SSM, the boundary enhancement of F_{local} and F_{sg} can be formulated as follows:

$$F_i^{\text{edge}} = \phi_{1\times1}\left(F_i - \text{Avg}\left(F_i\right)\right) \tag{9}$$

where $\phi_{1\times1}$ processes the input as Conv1 $\times$ 1 $\rightarrow$ BN $\rightarrow$ PReLU and Avg$(\cdot)$ denotes the average pooling layer with 3×3 kernel. Notably, in the SSM, $\phi_{1\times1}$ utilizes the PReLU function since we find that retaining negative values more effectively highlights local contrast cues. Then, the SSM utilizes the $F_{\text{sg}}^{\text{edge}}$ as guidance to enhance the structure representation of salient objects in $F_{\text{local}}^{\text{edge}}$. The output of SSM can be formulated as follows:

$$F_s = F_{\text{local}} + \phi_{1\times1}\left(F_{\text{local}}^{\text{edge}} \circ F_{\text{sg}}^{\text{edge}}\right) \tag{10}$$

where $\circ$ denotes the element-wise multiplication. Finally, SARD fuses the structure refined shallow feature F_s and

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high-level feature F_{sg} to generate output S_f, which can be formulated as follows:

$$S_f = \mathrm{Conv}_{1\times1}\left(\phi_{3\times3}\left(\phi_{3\times3}\left(\mathrm{Cat}\left[F_s, F_{sg}\right]\right)\right)\right). \tag{11}$$

The prediction of ORSI-SAM is a weighted fusion of SARD output S_f and mask decoder prediction S_m. The mask decoder from the original MobileSAM generates S_m through image embedding and learnable prompts, enabling precise object localization. The detail can be formulated as follows:

$$S_{\mathrm{out}} = w_1 \cdot S_f + w_2 \cdot \mathrm{Up}\left(S_m\right) \tag{12}$$

where w_1 and w_2 are learnable weights to adaptively fuse S_f and S_m for generating final result and Up(·) is utilized to upsample the S_m to match the input image size. Through an adaptive weighting mechanism, ORSI-SAM achieves optimal information complementarity between these two components. This design not only enhances boundary representation of salient objects in the final prediction but also improves localization accuracy.

D. Loss Function

In this work, we utilize the binary cross-entropy (BCE) loss, intersection over union (IOU) loss, and $L1$ loss to train our framework. The total loss consists of two parts to optimize the learnable prompt $S_{\mathrm{mask}} = \mathrm{Up}(S_g)$ and final prediction S_{out}. The total loss can be formulated as follows:

$$\mathcal{L}\left(S, S_{\mathrm{gt}}\right) = \mathcal{L}_{\mathrm{BCE}} + \mathcal{L}_{\mathrm{IOU}} + \mathcal{L}_{L1} \tag{13}$$

$$\mathcal{L}_{\mathrm{total}} = \mathcal{L}\left(S_{\mathrm{mask}}, S_{\mathrm{gt}}\right) + \mathcal{L}\left(S_{\mathrm{out}}, S_{\mathrm{gt}}\right) \tag{14}$$

where S_{gt} is the ground truth of salient objects.

IV. Experiments

A. Experiment Settings

1) Datesets: In this article, we conduct experiments on the ORSSD [9] and EORSSD [32], and ORSI-4199 [33] datasets. The ORSSD dataset is the first dataset specifically for the ORSI-SOD task, which has 600 images for training and 200 images for testing. The EORSSD dataset is an extended version of ORSSD, which adds 1200 ORSIs to the ORSSD dataset, resulting in 2000 ORSIs. This dataset contains 1400 images for training and 600 images for testing. ORSI-4199 contains 2000 training images and 2199 testing images. All three datasets utilize pixel-level annotations to identify salient objects for each ORSI.

2) Evaluation Metrics: Following previous methods [6], [10], we compare the proposed method with previous methods in terms of quantitative evaluation metrics and computational complexity. For quantitative evaluation metrics, S-measure (S_α, $\alpha = 0.5$) [72], F-measure (F_β, $\beta^2 = 0.3$) [73], E-measure (E_ξ) [74], and mean absolute error ($\mathcal{M}$). S_α assesses the similarity of object-part regions and entire objects. F_β is calculated as the weighted average of precision and recall. E_ξ measures the similarity between predictions and labels by considering local and image-level saliency values simultaneously. $\mathcal{M}$ measures the average absolute difference at all pixel locations. Computational complexity metrics include the inference speed measured in frames/s (FPS), the parameter amount (Param) measured in millions (M), and the number of FLOPs measured in giga (G). The inference speed is reported with a batch size of 1 and no I/O time. We evaluate FLOPs and FPS using an input size of 288×288 in the subsequent sections.

3) Implementation Details: Our experiments are implemented using PyTorch and conducted on a server equipped with two NVIDIA A100 GPUs. The training configurations and parameters are as follows: the input image size is set to 288×288 and data augmentation strategies include flipping and rotation. The MobileSAM is optimized using AdamW with a weight decay of $1e^{-4}$, and the batch size is set to 32. The initial learning rate is $5e^{-4}$, reduced using an exponential decay schedule. A warm-up period of 5 epochs is applied, and the model is trained for a total of 60 epochs. For initialization, the weights of the image encoder and mask decoder are preloaded from the MobileSAM model [22], while the remaining components of the proposed ORSI-SAM are randomly initialized. Consistent with standard practices, training and testing are conducted on separate datasets as specified in [10].

B. Comparison With State-of-the-Art Methods

We compare our proposed ORSI-SAM with 32 state-of-the-art SOD methods, comprising 11 conventional NSI-SOD methods (**CN**) and 12 conventional ORSI-SOD methods (**CR**). The CN includes DSS [24], RADF [52], R3Net [53], PoolNet [54], EGNet [25], GCPA [55], MINet [56], ITSD [57], GateNet [58], SUCA [59], and PA-KRN [60]. The CR includes LVNet [9], DAFNet [32], SARNet [37], MJRBM [33], EMFINet [5], ACCoNet [11], ERPNet [61], SFANet [35], WeightNet [62], HFCNet [36], LGIPNet [63], and MRBINet [64]. Additionally, four lightweight SOD methods (**LN**) and five lightweight ORSI-SOD methods (**LR**) are also utilized for evaluating the tradeoff between the accuracy and efficiency of our method. The LN includes CSNet [65], SAMNet [66], HVPNet [67], and ADMNet [68]. The LR includes CorrNet [6], SeaNet [10], MEANet [69], ISAANet [70], and RAMENet [71].

Moreover, we also compare our method with recent SAM-based methods, including SAM fine-tuning [15], SAM-Adapter [75], SAM-LoRA [76], and MDSAM [18]. Notably, MDSAM is an SAM-based model specifically tailored for the SOD task.

1) Comparison With State-of-the-Art SOD Methods for NSIs: As presented in Tables I and II, we compare the performance of our ORSI-SAM with recent state-of-the-art SOD methods for NSIs across three public datasets: ORSSD, EORSSD, and ORSI-4199. Our method demonstrates superior performance and significant improvements across all datasets. Notably, compared to the best-performing PA-KRN, ORSI-SAM reduces parameter size by 11.6 times while achieving substantial gains on all evaluation metrics. For instance, on the EORSSD dataset, PA-KRN achieves an S_α score of 0.9192, whereas ORSI-SAM improves this to 0.9418. High-performance NSI-SOD methods, such as GateNet and PA-KRN, generally demand extensive parameters and computational resources. While ITSD achieves a seven times

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TABLE I

QUANTITATIVE EVALUATION. THE EVALUATION METRICS INCLUDE: STRUCTURE MEASURE (S_m), MAXIMUM, MEAN, AND ADAPTIVE F-MEASURE ($F_\beta^{\max}$, F_β^{MEAN}, AND F_β^{ADA}), MAXIMUM, MEAN, ADAPTIVE E-MEASURE ($E_\xi^{\max}$, E_ξ^{MEAN}, AND E_ξ^{ADA}), AND MAE ($\mathcal{M}$). FOR MAE, SMALLER IS BETTER, WHILE FOR THE OTHER METRICS, LARGER IS BETTER. THE BEST THREE RESULTS ARE MARKED IN RED, BLUE, AND GREEN

Methods	Type	Input size	Params (M) ↓	FLOPs (G) ↓	EORSSD								ORSSD							
					$S_\alpha\uparrow$	$F_\beta^{max}\uparrow$	$F_\beta^{mean}\uparrow$	$F_\beta^{adp}\uparrow$	$E_\xi^{max}\uparrow$	$E_\xi^{mean}\uparrow$	$E_\xi^{adp}\uparrow$	$\mathcal{M}\downarrow$	$S_\alpha\uparrow$	$F_\beta^{max}\uparrow$	$F_\beta^{mean}\uparrow$	$F_\beta^{adp}\uparrow$	$E_\xi^{max}\uparrow$	$E_\xi^{mean}\uparrow$	$E_\xi^{adp}\uparrow$	$\mathcal{M}\downarrow$
DSS_{19} [24]	CN	400×300	62.2	114.6	.7868	.6849	.5801	.4597	.9186	.7631	.6933	.0186	.8262	.7467	.6962	.6206	.8860	.8362	.8085	.0206
$RADF_{18}$ [52]	CN	400×400	62.5	214.2	.8179	.7446	.6582	.4933	.9140	.8567	.7162	.0168	.8259	.7619	.6856	.5730	.9130	.8298	.7678	.0382
$R3Net_{18}$ [53]	CN	300×300	56.2	47.5	.8184	.7498	.6302	.4165	.9483	.8294	.6462	.0171	.8141	.7456	.7383	.7379	.8913	.8681	.8887	.0399
$PoolNet_{19}$ [54]	CN	400×300	53.6	123.4	.8207	.7545	.6406	.4611	.9292	.8193	.6836	.0210	.8403	.7706	.6999	.6166	.9343	.8650	.8124	.0358
$EGANet_{19}$ [25]	CN	∼380×320	108.1	291.9	.8601	.7880	.6967	.5379	.9570	.8775	.7566	.0110	.8721	.8332	.7500	.6452	.9731	.9013	.8226	.0216
$GCPA_{20}$ [55]	CN	320×320	67.1	54.3	.8869	.8347	.7905	.6723	.9524	.9167	.8647	.0102	.9026	.8687	.8433	.7861	.9509	.9341	.9205	.0168
$MINet_{20}$ [56]	CN	320×320	47.6	146.3	.9040	.8344	.8174	.7705	.9442	.9346	.9243	.0093	.9040	.8761	.8574	.8251	.9545	.9454	.9423	.0144
$ITSD_{20}$ [57]	CN	288×288	17.1	54.5	.9050	.8523	.8221	.7421	.9556	.9407	.9103	.0106	.9050	.8735	.8502	.8068	.9601	.9482	.9335	.0165
$GateNet_{20}$ [58]	CN	384×384	100.0	108.3	.9114	.8566	.8228	.7109	.9610	.9385	.8909	.0095	.9186	.8871	.8679	.8229	.9664	.9538	.9428	.0137
$SUCA_{21}$ [59]	CN	256×256	117.7	56.4	.8988	.8229	.7949	.7260	.9520	.9277	.9082	.0097	.8989	.8484	.8237	.7748	.9584	.9400	.9194	.0145
$PA\text{-}KRN_{21}$ [60]	CN	600×600	141.1	617.7	.9192	.8639	.8358	.7993	.9616	.9536	.9416	.0104	.9239	.8890	.8727	.8548	.9680	.9620	.9579	.0139
$LVNet_{19}$ [9]	CR	128×128	-	-	.8630	.7794	.7328	.6284	.9254	.8801	.8445	.0146	.8815	.8263	.7995	.7506	.9456	.9259	.9195	.0207
$DAFNet_{21}$ [32]	CR	128×128	29.4	68.5	.9166	.8614	.7845	.6427	.9861	.9291	.8446	.0060	.9191	.8928	.8511	.7876	.9771	.9539	.9360	.0113
$SARNet_{21}$ [37]	CR	336×336	25.9	129.7	.9240	.8719	.8541	.8304	.9620	.9555	.9536	.0099	.9134	.8850	.8619	.8512	.9557	.9477	.9464	.0187
$MJRBM_{22}$ [33]	CR	352×352	43.5	95.7	.9197	.8656	.8239	.7066	.9646	.9350	.8897	.0099	.9204	.8842	.8566	.8022	.9623	.9415	.9328	.0163
$EMFINet_{22}$ [5]	CR	256×256	107.3	480.9	.9290	.8720	.8486	.7984	.9711	.9604	.9501	.0084	.9366	.9002	.8856	.8617	.9737	.9671	.9663	.0109
$ACCoNet_{23}$ [11]	CR	256×256	127.0	51.3	.9290	.8837	.8552	.7969	.9727	.9653	.9450	.0074	.9437	.9149	.8971	.8806	.9796	.9754	.9721	.0080
$ERPNet_{23}$ [61]	CR	256×256	77.2	171.9	.9252	.8743	.8269	.7170	.9665	.9366	.8994	.0082	.9352	.9036	.8798	.8392	.9738	.9604	.9531	.0114
$SFANet_{24}$ [35]	CR	256×256	25.1	7.7	.9349	.8833	.8680	.8492	.9769	.9726	.9681	.0058	.9453	.9192	.9063	.8984	.9830	.9789	.9799	.0077
$WeightNet_{24}$ [62]	CR	256×256	66.3	21.2	.9417	.8908	.8779	.8521	.9809	.9749	.9670	.0060	.9512	.9278	.9125	.8979	.9864	.9823	.9776	.0075
$HFCNet_{24}$ [36]	CR	224×224	140.8	120.4	.9449	.8951	.8646	.8051	.9842	.9719	.9386	.0049	.9543	.9269	.9049	.8710	.9884	.9798	.9748	.0070
$LGIPNet_{25}$ [63]	CR	352×352	52.1	47.0	.9392	.8924	.8770	.8623	.9806	.9753	.9719	.0053	.9507	.9266	.9129	.9027	.9876	.9829	.9809	.0066
$MRBINet_{25}$ [64]	CR	256×256	32.4	42.8	.9351	.8852	.8768	.8626	.9766	.9721	.9729	.0056	.9474	.9199	.9110	.9058	.9851	.9812	.9793	.0069
$CSNet_{20}$ [65]	LN	224×224	0.1	0.7	.8364	.8341	.7656	.6319	.9535	.8929	.8339	.0169	.8910	.8790	.8285	.7615	.9628	.9179	.9068	.0186
$SAMNet_{21}$ [66]	LN	336×336	1.3	0.5	.8622	.7813	.7214	.6114	.9421	.8700	.8284	.0132	.8761	.8137	.7531	.6843	.9478	.8818	.8656	.0217
$HVPNet_{21}$ [67]	LN	336×336	1.2	1.1	.8734	.8036	.7377	.6202	.9482	.8721	.8270	.0110	.8610	.7938	.7396	.6726	.9320	.8717	.8471	.0225
$ADMNet_{24}$ [68]	LN	368×368	0.8	3.3	.8784	.7761	.7670	.7473	.9194	.9010	.9091	.0099	.9060	.8487	.8392	.8276	.9480	.9360	.9430	.0168
$CorrNet_{22}$ [6]	LR	256×256	4.1	21.1	.9289	.8778	.8620	.8311	.9696	.9646	.9593	.0083	.9380	.9129	.9002	.8875	.9790	.9746	.9721	.0098
$SeaNet_{23}$ [10]	LR	288×288	2.8	1.7	.9208	.8649	.8519	.8304	.9710	.9651	.9602	.0073	.9260	.8942	.8772	.8625	.9767	.9722	.9670	.0105
$MEANet_{24}$ [69]	LR	352×352	3.3	9.6	.9282	.8766	.8678	.8557	.9708	.9658	.9663	.0070	.9340	.9033	.8934	.8860	.9768	.9730	.9715	.0098
$ISAANet_{24}$ [70]	LR	352×352	3.6	4.3	.9351	.8869	.8748	.8643	.9761	.9717	.9734	.0063	.9429	.9166	.9034	.8982	.9836	.9787	.9777	.0077
$RAMENet_{25}$ [71]	LR	352×352	5.2	8.7	.9419	.8971	.8840	.8678	.9822	.9778	.9735	.0050	.9489	.9229	.9147	.9097	.9862	.9819	.9797	.0070
ORSI-SAM (Ours)	LR	288×288	12.2	8.9	.9418	.8952	.8787	.8565	.9816	.9771	.9723	.0041	.9527	.9299	.9166	.9066	.9898	.9862	.9844	.0062

* **CN:** Conventional NSI-SOD method; **LN:** Lightweight NSI-SOD method;
* **CR:** Conventional ORSI-SOD method; **LR:** Lightweight ORSI-SOD method;

TABLE II

QUANTITATIVE EVALUATION ON ORSI-4199. THE BEST THREE RESULTS ARE MARKED IN RED, BLUE, AND GREEN

Method	Type	ORSI-4199							
		$S_\alpha\uparrow$	$F_\beta^{max}\uparrow$	$F_\beta^{mean}\uparrow$	$F_\beta^{adp}\uparrow$	$E_\xi^{max}\uparrow$	$E_\xi^{mean}\uparrow$	$E_\xi^{adp}\uparrow$	$\mathcal{M}\downarrow$
$R3Net_{18}$ [53]	CN	.8142	.7847	.7790	.7776	.8880	.8722	.8645	.0401
$PoolNet_{19}$ [54]	CN	.8271	.8010	.7779	.7382	.8964	.8676	.8531	.0974
$EGANet_{19}$ [25]	CN	.8464	.8267	.8041	.7650	.9161	.8947	.8620	.0440
$PA\text{-}KRN_{21}$ [60]	CN	.8491	.8415	.8324	.8200	.9280	.9168	.9063	.0382
$MJRBM_{22}$ [33]	CR	.8593	.8493	.8309	.7995	.9311	.9102	.8891	.0374
$EMFINet_{22}$ [5]	CR	.8675	.8584	.8479	.8186	.9340	.9257	.9136	.0330
$ERPNet_{23}$ [61]	CR	.8670	.8584	.8479	.8186	.9340	.9257	.9136	.0357
$ACCoNet_{23}$ [11]	CR	.8775	.8686	.8620	.8581	.9412	.9342	.9167	.0314
$SFANet_{24}$ [35]	CR	.8761	.8710	.8659	.8647	.9447	.9385	.9202	.0292
$WeightNet_{24}$ [62]	CR	.8884	.8870	.8813	.8802	.9534	.9456	.9283	.0267
$HFCNet_{24}$ [36]	CR	.8827	.8831	.8718	.8322	.9525	.9433	.9246	.0281
$LGIPNet_{25}$ [63]	CR	.8854	.8791	.8668	.8268	.9499	.9432	.9213	.0276
$MRBINet_{25}$ [64]	CR	.8824	.8800	.8755	.8755	.9489	.9425	.9220	.0268
$CSNet_{20}$ [65]	LN	.8241	.8142	.7674	.7162	.9096	.8586	.8447	.0524
$SAMNet_{21}$ [66]	LN	.8409	.8249	.8029	.7744	.9186	.8938	.8781	.0432
$HVPNet_{21}$ [67]	LN	.8471	.8295	.8041	.7652	.9201	.8956	.8687	.0419
$ADMNet_{24}$ [68]	LN	.8488	.8337	.8292	.8287	.9195	.9123	.9190	.0361
$CorrNet_{22}$ [6]	LR	.8623	.8560	.8513	.8534	.9330	.9206	.9142	.0366
$SeaNet_{23}$ [10]	LR	.8722	.8653	.8591	.8556	.9426	.9363	.9197	.0308
$MEANet_{24}$ [69]	LR	.8708	.8662	.8624	.8638	.9417	.9338	.9207	.0312
$ISAANet_{24}$ [70]	LR	.8732	.8760	.8696	.8694	.9497	.9431	.9239	.0281
$RAMENet_{25}$ [71]	LR	.8821	.8839	.8779	.8753	.9539	.9490	.9512	.0262
ORSI-SAM	LR	.8848	.8822	.8777	.8777	.9525	.9471	.9238	.0257

reduction in parameter size and halves the FLOPs compared to these methods, its performance remains suboptimal. Furthermore, lightweight SOD methods designed for NSIs, such as SAMNet and HVPNet, prioritize efficiency but suffer significant performance degradation when applied to ORSI-SOD tasks. For example, HVPNet achieves only 0.8734 in S_α and 0.0110 in $\mathcal{M}$, which is considerably lower than the recent ORSI-SOD approaches. This highlights the limited generalization capability of lightweight models in complex remote sensing scenarios compared to NSIs, emphasizing the necessity of specialized methods for ORSI-SOD. Compared with the latest state-of-the-art methods, we observe that our method demonstrates superior performance over LGIPNet [63] and MRBINet [64] across all three benchmark datasets. For RAMENet [71], our approach achieves comparable results on the EORSSD and ORSI-4199 datasets while outperforming it on the ORSSD dataset. RAMENet was published on July 1 and can be regarded as the contemporaneous work.

2) Comparison With State-of-the-Art SOD Methods for ORSIs: For ORSI-SOD methods specifically designed for remote sensing imagery, ORSI-SAM demonstrates comparable performance to recent state-of-the-art approaches while maintaining a significantly lower parameter size and computational cost (as shown in Tables I and II). The best-performing method, HFCNet, leverages a hybrid encoder combining Swin Transformer and VGG to extract heterogeneous features for modeling the relationship between local information and global information. However, this performance advantage comes with the cost of 140 M parameters and 120 G FLOPs, making it impractical for deployment on resource-constrained satellite platforms. Recent lightweight ORSI-SOD methods, including CorrNet and SeaNet, effectively reduce computational overhead and model parameters, but their limited representative capacity results in a noticeable performance gap compared to more complex models. This tradeoff highlights

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TABLE III

COMPARISON WITH SAM-BASE METHODS ON ORSSD, EORSSD, AND ORSI-4199. RED DENOTES TRAINABLE PARAMETERS DURING MODEL TRAINING. THE BEST RESULTS ARE MARKED IN BOLD

Method	Params (M)	FLOPs (G)	ORSSD				EORSSD				ORSI-4199			
			$S_\alpha \uparrow$	$F_\beta^{mean} \uparrow$	$E_\xi^{mean} \uparrow$	$\mathcal{M} \downarrow$	$S_\alpha \uparrow$	$F_\beta^{mean} \uparrow$	$E_\xi^{mean} \uparrow$	$\mathcal{M} \downarrow$	$S_\alpha \uparrow$	$F_\beta^{mean} \uparrow$	$E_\xi^{mean} \uparrow$	$\mathcal{M} \downarrow$
SAM fine-tuning	90.23(4.29)	39.36	0.9095	0.8688	0.9584	0.0132	0.9138	0.8437	0.9613	0.0070	0.8570	0.8486	0.9297	0.0340
SAM-Adapter [77]	93.80 (7.86)	40.50	0.9468	0.9072	0.9806	0.0072	0.9387	0.8743	0.9756	0.0051	0.8791	0.8731	0.9435	0.0266
SAM-LoRA [76]	97.31(11.37)	43.82	0.9446	0.9072	0.9793	0.0071	0.9357	0.8697	0.9759	0.0051	0.8777	0.8732	0.9448	0.0269
MDSAM [18]	100.51(14.57)	46.22	0.9482	0.9104	0.9803	0.0063	0.9380	0.8740	0.9744	0.0050	0.8836	0.8759	0.9448	0.0268
MobileSAM fine-tuning	9.57(9.57)	4.39	0.9370	0.8924	0.9765	0.0080	0.9222	0.8471	0.9693	0.0064	0.8748	0.8648	0.9424	0.0279
MobileSAM-Adapter [77]	9.77(4.38)	4.47	0.9256	0.8710	0.9668	0.0100	0.9159	0.8343	0.9652	0.0075	0.8722	0.8618	0.9418	0.0282
MobileSAM-LoRA [76]	9.96(4.57)	4.66	0.9324	0.8820	0.9733	0.0094	0.9152	0.8346	0.9662	0.0065	0.8732	0.8619	0.9424	0.0289
MobileSAM-MDSAM [18]	12.97(7.78)	9.28	0.9437	0.9049	0.9809	0.0074	0.9327	0.8642	0.9727	0.0062	0.8819	0.8733	0.9461	0.0263
ORSI-SAM	12.16 (12.16)	8.87	**0.9527**	**0.9166**	**0.9862**	**0.0062**	**0.9418**	**0.8787**	**0.9771**	**0.0041**	**0.8848**	**0.8777**	**0.9471**	**0.0257**

the importance of designing efficient yet high-performing architectures for ORSI-SOD applications. Our ORSI-SAM achieves state-of-the-art performance in the EORSSD and ORSSD datasets while reducing the parameter count by 91% and the FLOPs by 93% compared to HFCNet. Compared to HFCNet, ORSI-SAM achieves superior performance across F_β, E_ξ, and $\mathcal{M}$ metrics (e.g., 0.0041 versus 0.0049 in $\mathcal{M}$ on EORSSD), demonstrating its effectiveness. Compared to CorrNet and SeaNet, ORSI-SAM increases the parameter size but achieves a more favorable tradeoff between accuracy and efficiency. The superior performance of ORSI-SAM highlights the benefits of leveraging pretraining knowledge in large-scale natural images for ORSI-SOD tasks, showcasing the potential of lightweight SAM architectures for remote sensing applications.

3) Comparison With SAM-Base SOD Methods: We further compare ORSI-SAM with recent SAM-based methods. Using ViT-base as the backbone for SAM, we explore various fine-tuning strategies, including decoder fine-tuning, Adapter [77], and LoRA [76]. Additionally, we evaluate the effectiveness of MDSAM in the ORSI-SOD task, which is the first SAM-based architecture tailored for SOD in NSIs. The results, illustrated in Table III, reveal that fine-tuning the original SAM delivers outstanding performance on ORSI-SOD tasks. For instance, the SAM-Adapter [77] achieves results comparable to state-of-the-art methods such as HFCNet [36]. These findings underscore the effectiveness of SAM in leveraging knowledge from large-scale NSIs to significantly enhance scene understanding and object perception in ORSIs. However, compared to SAM, MobileSAM demonstrates a notable decline in performance. Importantly, all parameters in MobileSAM are fine-tuned, as opposed to limiting training to the decoder under SAM fine-tuning settings. For example, with adapter fine-tuning, MobileSAM-Adapter shows a reduction of 0.0212 in S_α and 0.0028 in $\mathcal{M}$ on the ORSSD dataset compared to SAM-Adapter. This emphasizes the critical need for targeted optimization of MobileSAM for the ORSI-SOD task. Our proposed ORSI-SAM strikes an effective balance between efficiency and accuracy by addressing information loss during the embedding stage of MobileSAM and mitigating the suppression of object representations.

4) Computational Complexity Comparison: To evaluate the efficiency of our ORSI-SAM, we compare parameter size and FPS between our ORSI-SAM and recent state-of-the-art approaches, including HFCNet [36], MDSAM [18], SeaNet [10], CorrNet [6], and ERPNet [35]. FPS is measured using an input resolution of 288 × 288. The comparison can be observed in Fig. 7. Our ORSI-SAM achieves an exceptional balance between accuracy and efficiency. For parameter size, our ORSI-SAM comprises 12.16 M parameters, achieving a 91.4% reduction in parameter count compared to HFCNet [36] while improving F_β^{mean} by 0.0141. Although our method introduces an approximately 10 M increase in parameter count compared to other lightweight ORSI-SOD methods such as SeaNet [10] and CorrNet [6], it achieves a significant performance improvement. Furthermore, we evaluate the FPS on the NVIDIA A100 GPU. As shown in Fig. 7, our method achieves a processing speed of 61 FPS, which is 16 FPS slower than SeaNet [10]. However, considering the tradeoff between speed and performance, our approach remains advantageous for practical deployment scenarios.

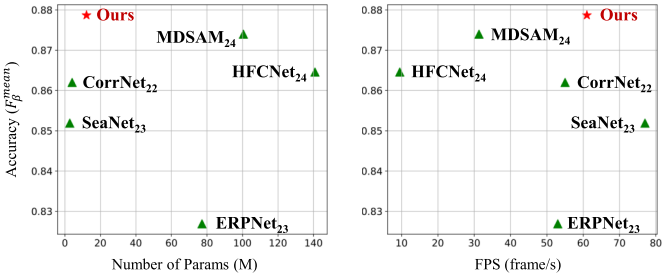


Fig. 7. Comparison of parameter size and FPS between our method and recent state-of-the-art approaches.

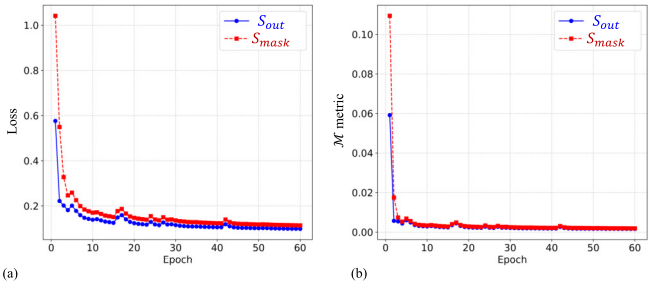


Fig. 8. Training loss and $\mathcal{M}$ curves of $L(S_{\text{out}}, S_{\text{gt}})$ and $L(S_{\text{mask}}, S_{\text{gt}})$ on the EORSSD dataset. (a) Training loss of prediction S_{out} and mask prompt S_{mask}. (b) $\mathcal{M}$ value of prediction S_{out} and mask prompt S_{mask}.

5) Training Analysis of ORSI-SAM: The optimization of our model is guided by a loss function computed between the ground truth S_{gt} and two components, prediction S_{out} and

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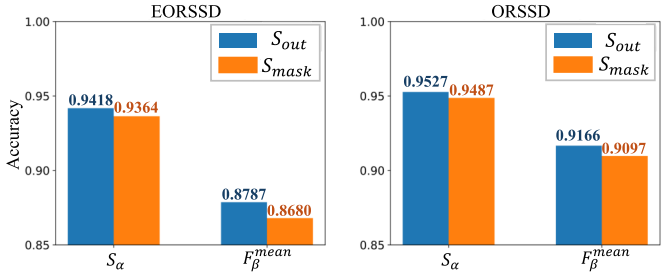


Fig. 9. Accuracy comparison of S_{out} and S_{mask} on EORSSD and ORSSD datasets. S_α and F_β^{mean} as utilized as metrics.

upsampled mask prompt S_{mask}. As shown in Fig. 8, we present the loss curves for prediction $\mathcal{L}(S_{\text{out}}, S_{\text{gt}})$ and $\mathcal{L}(S_{\text{mask}}, S_{\text{gt}})$ during training on the EORSSD dataset, as well as the $\mathcal{M}$ metric curve on the training set. We observe that the variation trends of the S_{out} and S_{mask} during training are consistent, indicating that the prediction accuracy is coupled with the learnable mask prompts generated by HIPG. The test accuracy of S_{out} and S_{mask} can be observed in Fig. 9. The performance of the saliency maps generated by the prompts S_{mask} is lower than that of the final output S_{out}, which highlights the decoder's ability to further refine the prompts and enhance the information. This finding motivates further exploration into generating high-quality prompts to enhance the final output performance in future work.

6) Qualitative Comparison: In Fig. 10, we qualitatively compare our ORSI-SAM with ten state-of-the-art ORSI-SOD methods across four challenging ORSI scenarios. In scenarios with large salient objects [Fig. 10(a)], lightweight methods like SeaNet and CorrNet tend to ignore specific salient regions. Meanwhile, recent approaches such as HFCNet and SFANet struggle with interference from shadows and similar background features, as observed in the second row of Fig. 10(a). Benefiting from enhanced suppression of background interference and strengthened object structural features, our method excels in object localization and accurate recognition. For scenarios containing multiple small objects [Fig. 10(b)], prior methods like SFANet and ACCoNet not only emphasize salient objects but also highlight redundant targets. In contrast, ORSI-SAM effectively suppresses redundant background information, leading to reduced false predictions. In cases involving irregular salient objects [Fig. 10(c)], our method effectively captures overall structures and produces smooth object edges, thanks to the SARD module's capability to model object structures. Scenarios with background-interfered salient objects [Fig. 10(d)] exemplify common challenges in ORSI. For instance, in the second column of Fig. 10(d), the low contrast between the river and its background poses challenges for recent methods such as SFANet and ACCoNet in distinguishing targets from their surroundings. While HFCNet can locate objects within similar backgrounds, it struggles to generate complete object structures. In contrast, ORSI-SAM effectively identifies low-contrast salient objects with textures and colors resembling their backgrounds, demonstrating its superior performance in handling low-contrast and background-interfered scenarios. This highlights the robustness of our method in addressing such challenges in ORSI-SOD.

TABLE IV
ABLATION STUDY OF EACH COMPONENT IN ORSI-SAM

Method	Params (M)	FLOPs (G)	EORSSD			
			$S_\alpha \uparrow$	$F_\beta^{mean} \uparrow$	$E_\xi^{mean} \uparrow$	$\mathcal{M} \downarrow$
Baseline	9.57	4.39	0.9222	0.8471	0.9693	0.0064
Baseline+HIPG	11.93	5.58	0.9296	0.8556	0.9717	0.0053
Baseline+SARD	9.81	7.68	0.9314	0.8614	0.9746	0.0051
ORSI-SAM	12.16	8.87	0.9418	0.8787	0.9771	0.0041

Additionally, we provide a qualitative comparison with SAM-based methods (including MDSAM, SAM-Adapter, SAM-LoRA, SAM, and MobileSAM), as illustrated in Fig. 11. Leveraging the pretrained knowledge from large-scale natural image datasets, SAM-based methods demonstrate strong capabilities in perceiving and localizing salient objects, particularly when fine-tuning the encoder of SAM. However, these approaches, including MDSAM and other fine-tuning methods, exhibit limitations in capturing detailed structures of targets, often resulting in partial information loss and unclear object boundaries. For instance, as depicted in the second row of Fig. 11, the tail section of the airplane is partially missing in the predictions of MDSAM, SAM-Adapter, and SAM-LoRA, highlighting their shortcomings in preserving fine structural details. In contrast, the saliency maps generated by our method effectively preserve the structural and detailed information of the objects, demonstrating its advantages in ORSI scenarios.

C. Model Analysis of ORSI-SAM

In this section, we present a comprehensive analysis to further evaluate our design, including the effectiveness of each proposed component, training strategies, ablation studies on HIPG and SARD, and performance under varying input sizes.

1) Ablation Study of Each Component in ORSI-SAM: In this section, we evaluate the effectiveness of the proposed components, as summarized in Table IV. Initially, we remove both the HIPG and SARD components and train the model with all parameters on the EORSSD dataset. This results in a notable performance decline, with S_α dropping from 0.9418 to 0.9222. Subsequently, we progressively introduce the proposed components. The results demonstrate the individual contribution of each module. Specifically, incorporating HIPG improves S_α by 0.0074, while adding SARD enhances S_α by 0.0092. These results highlight the effectiveness of the proposed components in reducing information loss and improving weak feature responses for salient objects.

As illustrated in Fig. 12, we provide qualitative comparisons for 12(a) w/o HIPG and (b) w/o SARD, respectively. In Fig. 12(a), w/o HIPG results in partial information loss in the predicted salient object. For example, as shown in the first row, certain ships are only partially detected, highlighting the loss of small object information during the downsampling process. Similarly, without SARD, the detailed information about objects, particularly their boundaries, becomes diminished. Additionally, low-contrast targets fail to be effectively identified. For example, in Fig. 12(b), the second row highlights

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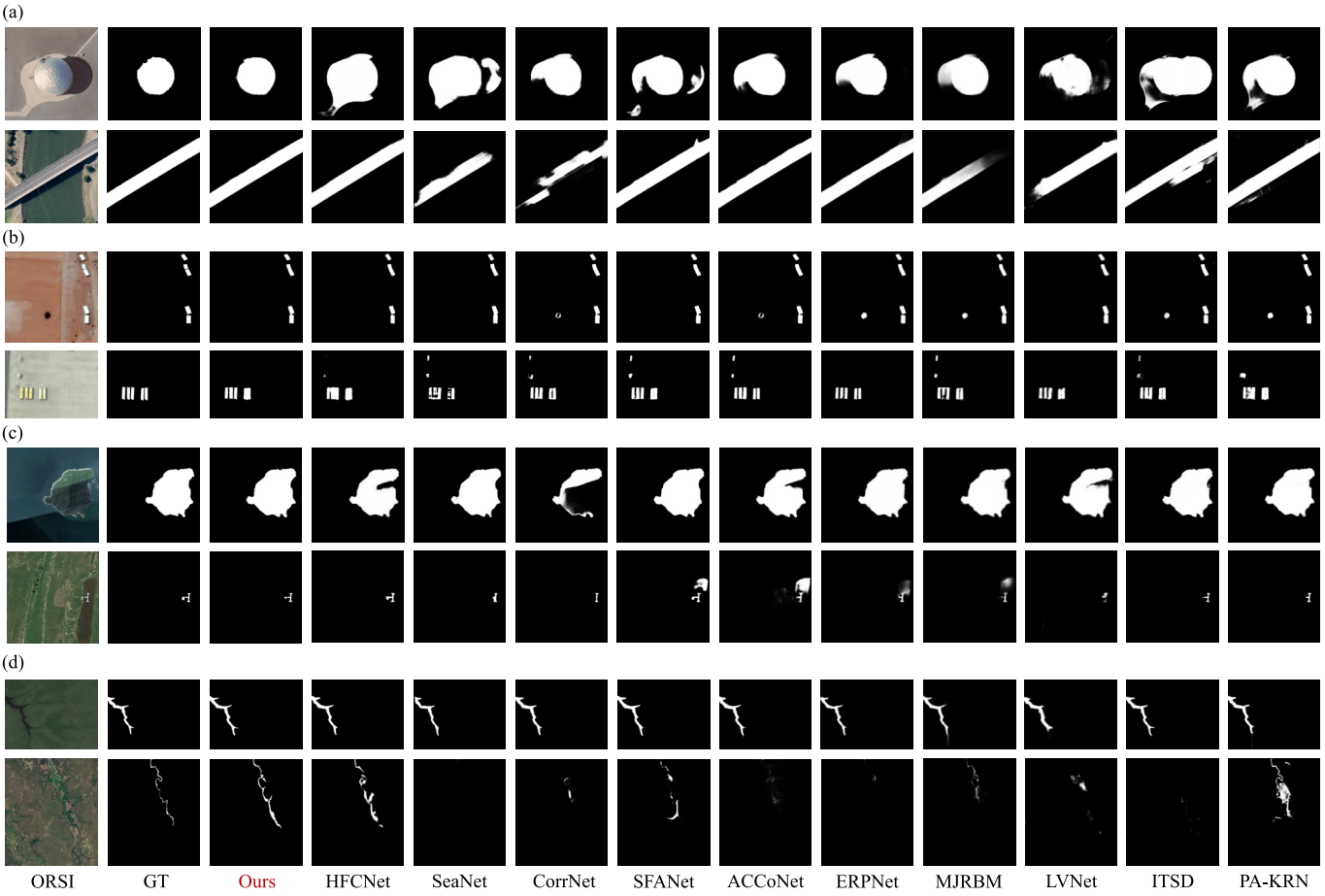


Fig. 10. Qualitative comparisons with ten representative state-of-the-art methods, including HFCNet [36], SeaNet [10], CorrNet [6], SFANet [35], AccoNet [11], ERPNet [61], MJRBM [33], LVNet [9], ITSD [57], and PA-KRN [60]. (a) Large salient regions. (b) Multiple small salient objects. (c) Salient objects with irregular topologies. (d) Salient objects with background interference.

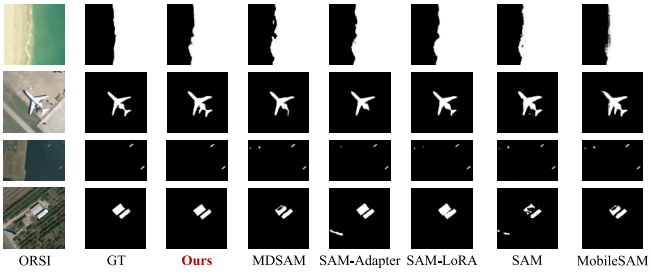


Fig. 11. Qualitative comparisons with five representative SAM-based methods, including MDSAM [18], SAM-Adapter [77], SAM-LoRA [76], SAM [15], and MobileSAM [22].

cases where low-contrast objects are not effectively identified against the background. These comparative experiments and visualizations demonstrate the effectiveness of our design for the ORSI-SOD task.

2) Ablation Study of Training Strategy on ORSI-SAM: In this section, we conduct different training strategies for ORSI-SAM to further explore the effectiveness of MobileSAM, including w/o pretrain weights, encoder frozen, LoRA, Adapter, and full training based on MDSAM.

As shown in the first row of Table V, excluding the pretrained weights of MobileSAM results in a significant performance drop on both ORSI-4199 and EORSSD datasets. For instance, the S_α score on ORSI-4199 decreases from 0.8848 to 0.8482, and on EORSSD, it drops from 0.9418 to 0.9056. This notable decline in accuracy underscores the critical importance of leveraging pretrained knowledge on the large-scale natural image dataset for enhancing ORSI-SOD performance.

Rows 2–4 of Table V present the evaluation results under different training strategies, including encoder frozen, LoRA, and Adapter approaches. When the encoder is frozen, ORSI-SAM achieves performance comparable to lightweight ORSI-SOD methods, such as CorrNet. This outcome further validates that the pretrained weights on natural images effectively capture semantic information in ORSI scenarios. Moreover, incorporating fine-tuning strategies for the encoder significantly enhances the model's generalization performance on ORSI datasets. Among these strategies, the adapter-based fine-tuning approach achieves superior results.

Finally, we compared the modules proposed in MDSAM [18] to validate the effectiveness of SARD and HIPG. For this, we replaced the backbone of MDSAM with MobileSAM and trained all parameters following the ORSI-SAM configuration. As shown in Table V (row 5), our method outperforms the MDSAM-based model (e.g., 0.8848 versus 0.8775 in S_α on

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TABLE V
ABLATION STUDY OF FINE-TUNING STRATEGIES ON ORSI-SAM

	Methods	Params (M)	ORSI-4199				EORSSD			
			$S_\alpha \uparrow$	$F_\beta^{mean} \uparrow$	$E_\xi^{mean} \uparrow$	$\mathcal{M} \downarrow$	$S_\alpha \uparrow$	$F_\beta^{mean} \uparrow$	$E_\xi^{mean} \uparrow$	$\mathcal{M} \downarrow$
ORSI-SAM	w/o Pretrain	12.16	0.8482	0.8292	0.9162	0.0398	0.9056	0.8311	0.9499	0.0096
	Encoder Forzen	12.16	0.8639	0.8573	0.9309	0.0341	0.9274	0.8596	0.9663	0.0064
	LoRA	12.55	0.8752	0.8693	0.9391	0.0293	0.9298	0.8619	0.9664	0.0066
	Adapter	12.36	0.8804	0.8716	0.9434	0.0270	0.9325	0.8635	0.9683	0.0050
	MDSAM	12.26	0.8775	0.8708	0.9407	0.0291	0.9393	0.8719	0.9748	0.0047
	Ours	12.16	**0.8848**	**0.8777**	**0.9471**	**0.0257**	**0.9418**	**0.8787**	**0.9770**	**0.0041**

TABLE VI
ABLATION STUDY ON THE PROPOSED HIPG

	Module		Params (M)	FLOPs (G)	EORRSD				ORSSD			
	LMFFM	SOPG			$S_\alpha \uparrow$	$F_\beta^{mean} \uparrow$	$E_\xi^{mean} \uparrow$	$\mathcal{M} \downarrow$	$S_\alpha \uparrow$	$F_\beta^{mean} \uparrow$	$E_\xi^{mean} \uparrow$	$\mathcal{M} \downarrow$
HIPG	✘	✘	9.81	7.68	0.9314	0.8614	0.9746	0.0051	0.9421	0.9018	0.9823	0.0076
	✔	✘	12.13	8.75	0.9394	0.8777	0.9758	0.0046	0.9512	0.9134	0.9805	0.0064
	✘	✔	9.83	8.09	0.9398	0.8734	0.9752	0.0048	0.9471	0.9060	0.9805	0.0065
	✔	✔	12.16	8.87	0.9418	0.8787	0.9771	0.0041	0.9527	0.9166	0.9862	0.0062

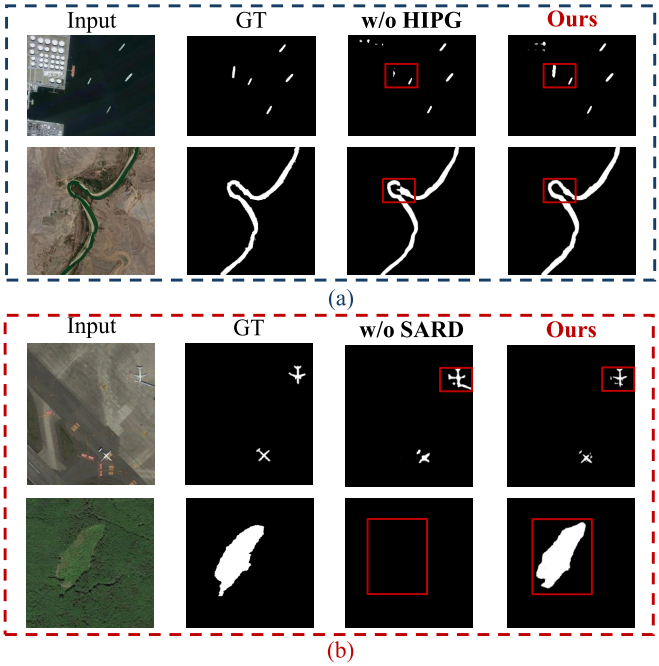


Fig. 12. Qualitative comparisons with (a) w/o HIPG and (b) w/o SARD. The red box indicates the false prediction part.

ORSI-4199), further illustrating the superiority of our method on the ORSI-SOD task.

3) Ablation Study on HIPG: As shown in Table VI, we present a detailed analysis of the proposed HIPG. The results indicate that removing either SOPG or LMFFM leads to a decline in performance. Specifically, w/o SOPG, the decoder lacks guidance from horizontal feature interactions, resulting in decreased accuracy. For example, the evaluation metric E_ξ^{mean} drops from 0.9862 to 0.9805 on ORSSD and from 0.9771 to 0.9758 on EORSSD. Similarly, w/o LMFFM disrupts the fusion of multiscale information, causing the loss of salient object features. Although SOPG partially compensates by providing shallow feature information, the removal of LMFFM still hampers the efficient representation of image embeddings, leading to further accuracy degradation. For instance, the evaluation metric F_β^{mean} decreases from 0.9166 to 0.9060 on ORSSD and from 0.8787 to 0.8734 on EORSSD. These findings underscore the significance and effectiveness of each component in HIPG.

4) Ablation Study on SARD: In this section, we perform an ablation study to evaluate the effectiveness of the proposed SARD module. As illustrated in Fig. 13, we design several variations of SARD to analyze the impact of our modifications. The details of these variations are outlined as follows.

1) *w/o Edge Enhancer:* As illustrated in Fig. 13(a), we exclude the edge enhancer and rely solely on high-level semantic features for redundancy reduction. This variation is designed to assess the impact of refining the structural representation of objects.
2) *w/o Gate:* As illustrated in Fig. 13(b), we exclude the guidance from high-level features. This design omits the filtering of redundant information in shallow features, allowing an evaluation of the impact of redundancy on the decoder's performance.
3) *Decoupled Supervision:* As shown in Fig. 13(d), our method combines the prediction S_m and S_f of the SAM decoder and SARD to generate the final saliency map. In contrast to this design, the decoupled supervision strategy applies independent supervision to both outputs using the ground truth, with only S_f utilized for inference.

The comparative performance results are presented in Table VII. We can find that: 1) w/o the edge enhancer and 2) w/o the gate demonstrate a noticeable decline in performance, particularly on the ORSSD dataset. For instance, the S_α metric decreases from 0.9527 to 0.9486 when the edge enhancer is removed and to 0.9488 when the gate guidance is omitted. This decline in accuracy highlights the critical role of refining structural representations and mitigating redundancy in achieving superior performance for ORSI-SOD tasks. Furthermore, a comparison between decoupled supervision and our weighted fusion approach reveals that our design

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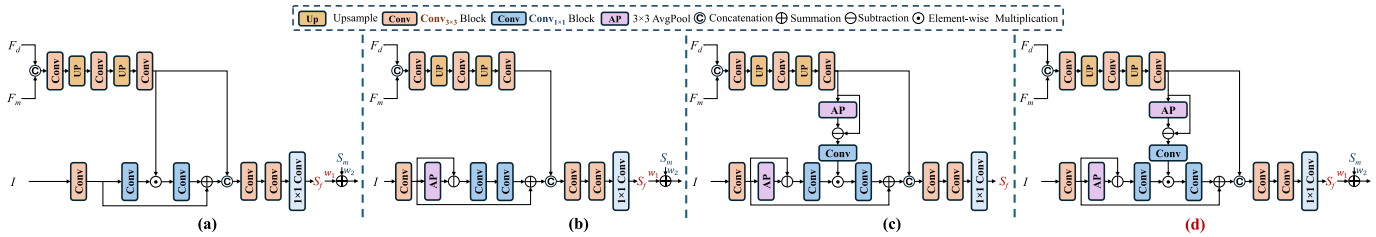


Fig. 13. Comparison of the SARD variations. (a) w/o edge enhancer, (b) w/o gate, (c) decoupled supervision, and (d) our SARD.

TABLE VII
ABLATION STUDY OF SARD. WE COMPARE FOUR VARIANTS OF SARD TO EVALUATE THE EFFECTIVENESS

	Module	Params (M)	FLOPs (G)	EORSSD				ORSSD			
				$S_\alpha \uparrow$	$F_\beta^{mean} \uparrow$	$E_\xi^{mean} \uparrow$	$\mathcal{M} \downarrow$	$S_\alpha \uparrow$	$F_\beta^{mean} \uparrow$	$E_\xi^{mean} \uparrow$	$\mathcal{M} \downarrow$
SARD	w/o Edge	12.15	8.70	0.9393	0.8729	0.9753	0.0044	0.9486	0.9101	0.9799	0.0069
	w/o Gate	12.16	8.70	0.9417	0.8775	0.9778	0.0044	0.9488	0.9096	0.9801	0.0070
	decouple	12.16	8.87	0.9416	0.8781	**0.9773**	0.0042	0.9510	0.9102	0.9812	0.0068
	fusion	12.16	8.87	**0.9418**	**0.8787**	0.9771	**0.0041**	**0.9527**	**0.9166**	**0.9862**	**0.0062**

TABLE VIII
EVALUATION OF ORSI-SAM ON VARIOUS INPUT SIZES

Methods	Input Size	FLOPs (G)	EORSSD			
			$S_\alpha \uparrow$	$F_\beta^{mean} \uparrow$	$E_\xi^{mean} \uparrow$	$\mathcal{M} \downarrow$
ORSI-SAM	224×224	4.85	0.9365	0.8707	0.9742	0.0052
	288×288	8.87	0.9418	0.8787	0.9771	**0.0041**
	384×384	14.85	0.9411	0.8834	**0.9778**	0.0051
	512×512	26.92	**0.9451**	**0.8873**	0.9758	0.0052

TABLE IX
EVALUATION OF ORSI-SAM ON THE SAM ENCODER

Method	EORSSD				ORSSD			
	$S_\alpha \uparrow$	$F_\beta^{mean} \uparrow$	$E_\xi^{mean} \uparrow$	$\mathcal{M} \downarrow$	$S_\alpha \uparrow$	$F_\beta^{mean} \uparrow$	$E_\xi^{mean} \uparrow$	$\mathcal{M} \downarrow$
Freeze	.9205	.8486	.9556	.0072	.9306	.8864	.9680	.0096
LoRA	.9406	**.8799**	**.9776**	.0050	.9502	.9115	.9828	.0069
Adapter	.9403	.8766	.9763	.0047	**.9535**	**.9167**	.9857	.0068
LMSA	.9410	.8773	.9766	**.0041**	.9485	.9105	.9816	.0065
ORSI-SAM	**.9418**	.8787	.9771	**.0041**	.9527	.9166	**.9862**	**.0062**

outperforms solely using S_f as the final prediction. For example, the S_α score improves from 0.9510 to 0.9527 on the ORSSD dataset, demonstrating the effectiveness of the weighted prediction fusion strategy.

5) Ablation Study on Input Size: In this section, we conduct the ablation study on the input image scale. In addition to the default input size of 288 × 288, we also evaluated the performance of our method at three additional input resolutions: 224 × 224, 384 × 384, and 512 × 512. As shown in Table VIII, a significant performance degradation is observed when the input resolution is reduced to 224 × 224. This decline can be attributed to the downsampling process in the MobileSAM backbone, which results in excessively low resolutions for subsequent encoder operations, leading to substantial information loss and adversely affecting the quality of the generated saliency maps. Increasing the input resolution leads to performance improvements in certain metrics. For instance, when the input resolution is increased to 512 × 512, the S_α score improves from 0.9418 to 0.9451. This observation motivates future exploration of ORSI-SOD frameworks tailored for high-resolution inputs.

D. Evaluation on Full SAM Backbones

In this section, we first explore whether the full version of SAM could further push the performance limits. We replace the MobileSAM encoder with the ViT-base from SAM in our ORSI-SAM. Due to the large number of parameters in the original SAM, we utilize fine-tuning strategies to train the model, including the freeze encoder, LoRA, Adapter, and LMSA. As shown in Table IX, we can see that the use of fine-tuning strategies can achieve a performance comparable to that of ORSI-SAM trained with all parameters. Therefore, we illustrate that SAM can further improve performance, but specific fine-tuning modules and strategies must be designed.

V. LIMITATION AND FUTURE WORK

Despite the significant advancements achieved by ORSI-SAM in the ORSI-SOD task, several limitations remain: 1) the model's reliance on pretrained knowledge from SAM may impede its adaptability to remote sensing scenes with substantial domain shifts; 2) the interaction module between the encoder and decoder introduces computational overhead, considerably impacting inference speed; and 3) effectively integrating pretrained knowledge from natural images into other remote sensing modalities (e.g., SAR images, infrared images, and hyperspectral images) remains an open challenge that requires further exploration. Building upon the current limitations, our future research will focus on the following key directions.

1) *Efficient Single-Branch Architecture:* Developing a lightweight yet high-performance single-branch framework for ORSI-SOD to optimize inference speed without compromising detection accuracy.
2) *Prior Knowledge Integration:* Investigating advanced methodologies to effectively incorporate domain-specific prior knowledge from ORSI scenes, aiming to

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improve target feature representation and discriminative capability.

3) *High-Resolution ORSI Processing:* Designing scalable detection architectures tailored for high-resolution ORSIs, addressing computational challenges posed by advancements in imaging technology.
4) *Multimodal Representation:* Exploring ORSI-SOD in multimodal contexts by aggregating saliency cues from heterogeneous data sources (e.g., SAR, optical, and hyperspectral imagery) to enhance scene understanding through cross-modal feature synergy.
5) *Adaptive Online Training:* Developing efficient online learning strategies to enhance model adaptability in dynamic environments, particularly for unseen scenarios and inputs corrupted by environmental disturbances (e.g., noise, occlusion, and atmospheric effects).

VI. Conclusion

In this study, we investigate leveraging the pretrained knowledge from large-scale natural image datasets to enhance ORSI-SOD. We propose a lightweight ORSI-SOD framework based on the SAM, referred to as ORSI-SAM. ORSI-SAM utilizes MobileSAM as the backbone to achieve a promising balance between accuracy and efficiency. To address the loss of object information inherent in lightweight SAM, we introduce the HIPG. HIPG enhances feature representation through horizontal feature aggregation and generates learnable mask prompts to assist the decoder in producing high-quality saliency maps. Additionally, we develop the SARD to strengthen the representation of object structural details suppressed by background interference in ORSIs, thereby improving the fine-grained detail and accuracy of segmented salient objects. Extensive experiments demonstrate that the proposed ORSI-SAM achieves performance comparable to the latest state-of-the-art ORSI-SOD methods while substantially reducing parameter size and computational complexity.

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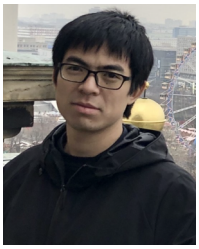
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